

Problem

High-fidelity scientific simulations are expensive, making exhaustive exploration of even modest parameter spaces impractical. This work asks a more useful question: *given a fixed budget of expensive solver calls, how should those evaluations be allocated to recover a rare scientific regime?*

The test environment is a five-dimensional injector/operating **combustion-stability analog**. Each sampled point is evaluated using a two-dimensional laminar OpenFOAM mixer. CFD-derived quantities, including mixing delay and spatial overlap, are then passed through a declared closed–closed one-dimensional Rayleigh-style map,

$$\sigma_{\text{analog}} = f(\text{mixing delay, overlap, } \dots),$$

where $\sigma_{\text{analog}} > 0$ denotes analog instability. The environment is deliberately a controlled analog rather than a predictive model of a flight engine.

AI-Guided Search

Every OpenFOAM run is treated as an expensive experiment. From the completed runs, an agent fits a Gaussian-process surrogate for σ_{analog} across the five-dimensional input domain.

The policy has two objectives. First, it searches uncertain or poorly characterized regions for a **missing regime**. Once instability is observed, it shifts toward **straddling the boundary** $\sigma_{\text{analog}} = 0$, selecting cases expected to improve knowledge of the stable/unstable transition.

AI-guided sampling is compared against Latin hypercube sampling (LHS) and uniform random sampling using identical parameter bounds, solver budgets, and evaluation data. Performance is measured on held-out cases rather than the adaptively selected training points.

The main metric is **unstable recall**: the fraction of the rare unstable regime successfully recovered under the limited solver budget.

Primary Result

Eight independent live OpenFOAM campaigns were performed with a budget of 16 expensive evaluations per campaign.

	Recall	Precision	F1
AI	0.68	0.86	0.76
LHS	0.33	0.85	0.46

AI achieved higher unstable recall in **7 of 8 seeds**. Its mean precision remained nearly identical to LHS, indicating that the increased recall did not arise simply from broadly labeling cases unstable. Because $n = 8$ is small, these values are reported as descriptive evidence; no statistical-significance claim is made.

A later frozen 16-seed live expansion, performed after development without retuning the policy, produced mean recall of **0.65 AI vs. 0.50**

LHS vs. 0.49 random, with AI exceeding LHS recall in 11 of 16 seeds. This is retained as additional post-development evidence; the eight-seed experiment remains the primary study.

Scientific Discovery

Live campaign seed 14 independently encountered analog-unstable points at both low and high values of the pattern-class parameter g . This unexpected observation motivated a controlled one-dimensional g -sweep at fixed low swirl.

The sweep revealed **two unstable intervals** along that slice. This is a narrower claim than discovering two disconnected five-dimensional regions: the adaptive campaign identified evidence that motivated the follow-up experiment, while the sweep established the two-interval structure only on the controlled one-dimensional cut.

Negative Results and V&V

The retained failures are central to the study. Seed 35 reverses the primary trend: LHS recall is 0.56 versus AI recall of 0.44. A later frozen one-shot hold-out using seed 101 similarly gives LHS recall of 0.76 versus AI recall of 0.59, although the AI campaign finds more unstable samples during exploration.

The high- g interval is also sensitive to modeling assumptions:

- it disappears on a coarser CFD mesh;
- it disappears under an $\omega + 10\%$ perturbation;
- it disappears for $V_{\text{crit}} = 0.035$;
- near-threshold unlike-impinging cases can flip labels when the solver is capped at 40 iterations.

These checks bound the scientific interpretation. The result demonstrates the behavior of the declared analog and the utility of the search protocol, not a universal physical instability boundary.

Reproducibility

The repository preserves the committed atlas, hold-out set, live campaign logs, verification and validation results, and frozen final-validation set. Installation is supported through a conventional venv/pip workflow or uv.

The command `cswe reproduce` skips live CFD when OpenFOAM is absent rather than inventing solver output. Historical JSONL campaign records are not rewritten.

Broader Significance

The reusable contribution is a protocol for scientific exploration: **freeze the analog and policy; impose a matched expensive-solver budget; use AI to choose the next experiment; evaluate on untouched data; follow discoveries with controlled tests; and retain the failures.**

The AI does not replace the simulator or governing equations. Its role is to determine *which expensive simulation should be run next*.